PSEUDO-COHERENCE IN A DISTINGUISHED TRIANGLE, WITH ONE DEGREE OF SLACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full, with the author's own remark that some details of representing the maps by complexes are omitted; the construction of the cone, the morphism of triangles and the comparison of the long exact sequences are all given.

Editorial note. The label lemma-cone-pseudo-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. This card belongs to the priority queue of the D-72 twin ruling (the sites-side statement is the designated counting side). The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Lifting a map of complexes through a quasi-isomorphism. Same chapter, the lemma allowing a map in the derived category to be represented by a map of complexes after refining a covering. Supplies the maps gamma_i of complexes used to build the cone.
- Strict perfectness of the cone on a morphism of strictly perfect complexes. Same chapter, the lemma stating that the cone on a morphism of strictly perfect complexes is strictly perfect. Supplies strict perfectness of C(gamma_i).
- The four and five lemmas on long exact cohomology sequences, and locality of pseudo-coherence. Homology, the four and five lemmas, and Same chapter, the locality lemma for pseudo-coherence. Supplies the conclusion that the induced map on the cone has the required cohomological behaviour, hence that M is m-pseudo-coherent.

Source reference. Original lemma 05QB: sources/stacks-chow/tags.txt, source lines 7370–7370. Prerequisite pointer invoked at proof line(s) 12226; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05QA: sources/stacks-chow/tags.txt, source lines 7369–7369. Prerequisite pointer invoked at proof line(s) 12225; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 08C5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11735–11735. Prerequisite pointer invoked at proof line(s) 12216; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08C8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11848–11848. Prerequisite pointer invoked at proof line(s) 12202; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08CC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12066–12066. Prerequisite pointer invoked at proof line(s) 12230; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site and $m \in \mathbf{Z}$. Let (K, L, M, f, g, h) be a distinguished triangle in $D(\mathcal{O})$.*

1. *If K is $(m+1)$-pseudo-coherent and L is m-pseudo-coherent then M is m-pseudo-coherent.*
2. *If K and M are m-pseudo-coherent, then L is m-pseudo-coherent.*
3. *If L is $(m+1)$-pseudo-coherent and M is m-pseudo-coherent, then K is $(m+1)$-pseudo-coherent.*

Proof. Proof of (1). Let U be an object of $\mathcal{C}$. Choose a covering $\{U_i \to U\}$ and maps $\alpha_i : \mathcal{K}_i^\bullet \to K|_{U_i}$ in $D(\mathcal{O}_{U_i})$ with $\mathcal{K}_i^\bullet$ strictly perfect and $H^j(\alpha_i)$ isomorphisms for $j > m+1$ and surjective for $j = m+1$. We may replace $\mathcal{K}_i^\bullet$ by $\sigma_{\geq m+1}\mathcal{K}_i^\bullet$ and hence we may assume that $\mathcal{K}_i^j = 0$ for $j < m+1$. After refining the covering we may choose maps $\beta_i : \mathcal{L}_i^\bullet \to L|_{U_i}$ in $D(\mathcal{O}_{U_i})$ with $\mathcal{L}_i^\bullet$ strictly perfect such that $H^j(\beta)$ is an isomorphism for $j > m$ and surjective for $j = m$. By Lemma 08C8 we can, after refining the covering, find maps of complexes $\gamma_i : \mathcal{K}^\bullet \to \mathcal{L}^\bullet$ such that the diagrams

$$\begin{array}{ccc} K|_{U_i} & \longrightarrow & L|_{U_i} \\ {\scriptstyle \alpha_i}\uparrow & & \uparrow{\scriptstyle \beta_i} \\ \mathcal{K}_i^\bullet & \xrightarrow{\gamma_i} & \mathcal{L}_i^\bullet \end{array}$$

are commutative in $D(\mathcal{O}_{U_i})$ (this requires representing the maps α_i, β_i and $K|_{U_i} \to L|_{U_i}$ by actual maps of complexes; some details omitted). The cone $C(\gamma_i)^\bullet$ is strictly perfect (Lemma 08C5). The commutativity of the diagram implies that there exists a morphism of distinguished triangles

$$(\mathcal{K}_i^\bullet, \mathcal{L}_i^\bullet, C(\gamma_i)^\bullet) \longrightarrow (K|_{U_i}, L|_{U_i}, M|_{U_i}).$$

It follows from the induced map on long exact cohomology sequences and Homology, Lemmas 05QA and 05QB that $C(\gamma_i)^\bullet \to M|_{U_i}$ induces an isomorphism on cohomology in degrees $> m$ and a surjection in degree m. Hence M is m-pseudo-coherent by Lemma 08CC.

Assertions (2) and (3) follow from (1) by rotating the distinguished triangle. □